

Assignment 1

1 Required Exercises

Exercise 1

When a C compiler compiles the following statement, how many tokens will it generate? [5points]

`int a3 = a * 3;`

When compiling the given statement, 7 tokens will be generated. The 1st token is the keyword `int`, the 2nd token is the identifier `a3`, the 3rd token is the operator `=`, the 4th token is the identifier `a`, the 5th token is the operator `*`, the 6th token is the integer literal `3`, and the 7th token is the delimiter.

Exercise 2

In a string of length n ($n > 0$), how many of the following are there?

在长度为 n ($n > 0$) 的字符串中，以下每项有多少？

1. Prefixes [5 points] 前缀[5 分]
2. Proper prefixes [5 points] 真前缀[5 分]
3. Prefixes of length m ($0 < m \leq n$) [5 points] 长度为 m 的前缀($0 < m \leq n$) [5 分]
4. Suffixes of length m ($0 < m \leq n$) [5 points] 长度为 m 的后缀($0 < m \leq n$) [5 分]
5. Proper prefixes of length m ($0 < m \leq n$) [10 points] 长度为 m 的真前缀($0 < m \leq n$) [10 分]
6. Substrings [10 points] 子串[10 分]
7. Subsequences [10 points] 子序列[10 分]

In a string of length n , there are:

1. $n + 1$ prefixes
2. $n - 1$ proper prefixes
3. 1 prefix of length m
4. 1 suffix of length m
5. 1 (if $m \neq n$) or 0 (if $m = n$) proper prefix of length m
6. $\frac{n(n+1)}{2} + 1$ substrings
7. 2^n subsequences

Exercise 3

Describe the languages denoted by the following regular expressions:

1. $((\epsilon|a)^*b^*)^*$ [5 points]
 2. $(a|b)^*a(a|b)(a|b)$ [5 points]
 3. $a^*ba^*ba^*ba^*$ [5 points]
1. The language of all strings over the alphabet $\{a, b\}$.
 2. The language of all strings over the alphabet $\{a, b\}$ that end with $\{aaa, aab, aba, abb\}$.
 3. The language of all strings over the alphabet $\{a, b\}$ where character b appears exactly 3 times.

Exercise 4

Write regular definitions or regular expressions for the following languages.

1. All strings representing valid telephone numbers in Shenzhen. A valid telephone number contains the country code (86), a hyphen, the area code 010, another hyphen, and eight digits where the first one cannot be zero (e.g., 86-010-62282045). [10 points]

2. All strings of a's and b's that start with a and end with b. [10 points]

3. All strings of lowercase letters that contain the five vowels in order. [10 points]

为以下语言编写正规定义或正则表达式。

1. 表示深圳有效电话号码的所有字符串。有效的电话号码包含国家代码（86）、一个连字符、区号 010、另一个连字符，以及八位数字且第一位数字不能为零（例如：86-010-62282045）。[10 分]

2. 所有以 a 开头并以 b 结尾的 a 和 b 的字符串。[10 分]

3. 所有包含按顺序出现的五个元音字母的小写字母字符串。[10 分]

1. $86 - 010 - [1 - 9][0 - 9]^7$

2. $a(a|b)^*b$

3. Let *consonant* $\rightarrow (b|c|d|f|g|h|j|k|l|m|n|p|q|r|s|t|v|w|x|y|z)^*$, the regular expression is *consonant a consonant e consonant i consonant o consonant u consonant*.

2 Optional Exercises

Exercise 1

Suppose we have a alphabet $\Sigma = a, b, c$, write regular definitions to describe all strings over Σ without repeated letters. [Hint: You may draw an NFA for the language and convert the NFA to regular definitions.]

假设我们有一个字母表 $\Sigma = \{a, b, c\}$ ，请写出正则表达式来描述 Σ 上所有没有重复字母的字符串。[提示：你可以先画出该语言的 NFA，然后将 NFA 转换为正则表达式。]

The DFA for the language of all strings over Σ without repeated letters is shown in fig. 1

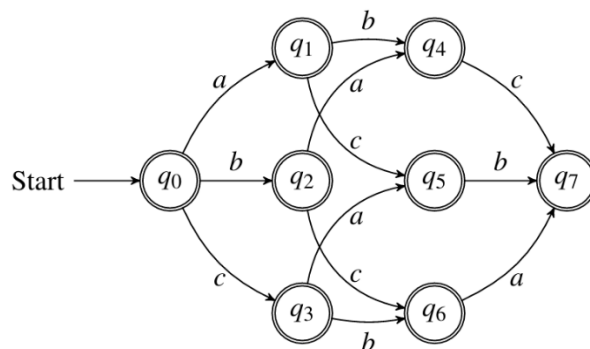


Figure 1: The DFA's transition diagram

According to the transition diagram, the regular expression for the language of all strings over Σ without repeated letters is

$(\epsilon|a)(\epsilon|b)(\epsilon|c)|$
 $(\epsilon|a)(\epsilon|c)(\epsilon|b)|$
 $(\epsilon|b)(\epsilon|a)(\epsilon|c)|$
 $(\epsilon|b)(\epsilon|c)(\epsilon|a)|$
 $(\epsilon|c)(\epsilon|a)(\epsilon|b)|$
 $(\epsilon|c)(\epsilon|b)(\epsilon|a).$

Assignment 2

1 Required Exercises

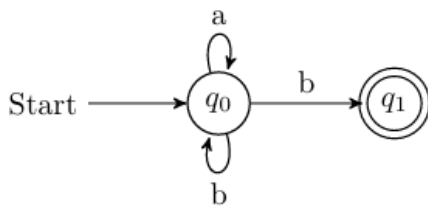
Exercise 1:

Design NFAs and DFAs to recognize each of the following regular languages. Detailed steps can be omitted.

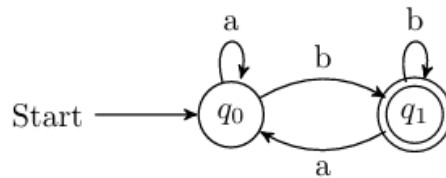
设计 NFA 和 DFA 来识别以下每个正则语言。可以省略详细步骤。

1. $L((a|b)^*b)$

(1) NFA

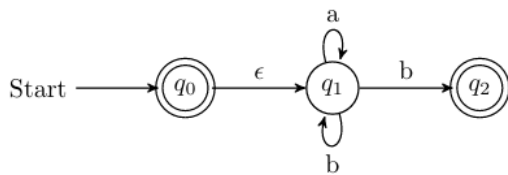


(2) DFA

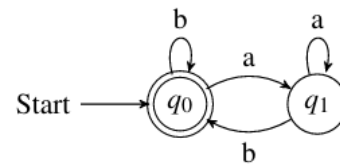


2. $L(((\epsilon|a)^*b)^*)$

(1) NFA

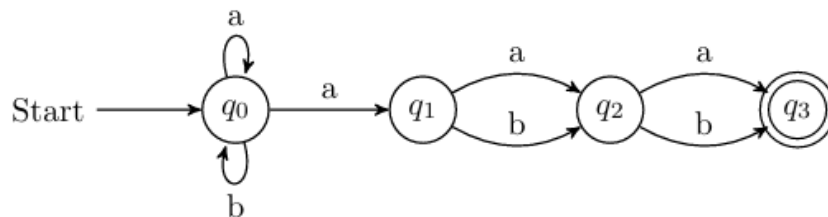


(2) DFA

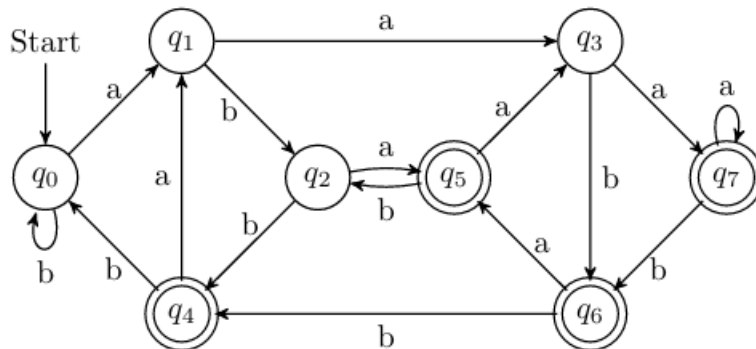


3. $L((a|b)^*a(a|b)(a|b))$

(1) NFA

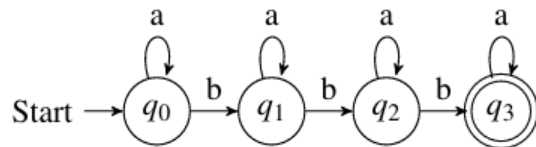


(2) DFA

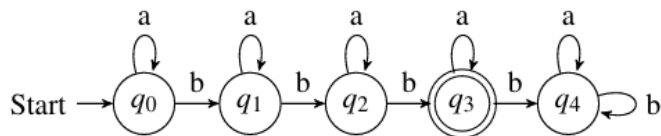


4. $L(a^*ba^*ba^*ba^*)$

(1) NFA



(2) DFA



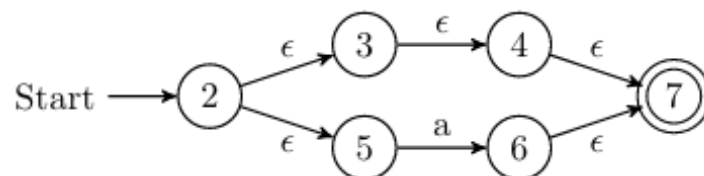
Exercise 2

Convert the following regular expressions to NFAs using the Thompson's Construction Algorithm (Algorithm 3.23 in the dragon book). Please put down the detailed steps.

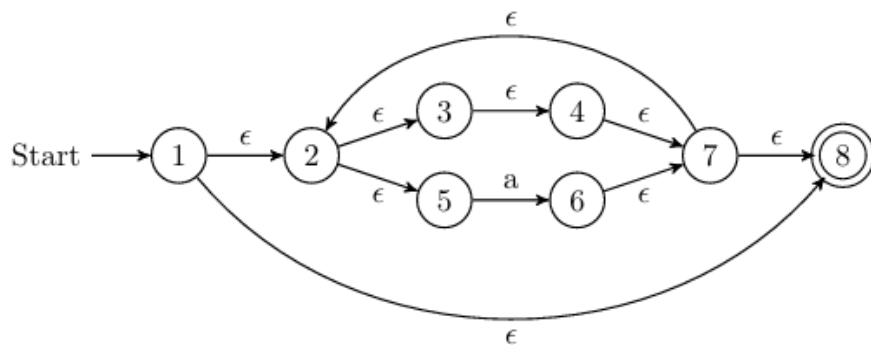
请使用汤普森构造算法（龙书中的算法 3.23）将以下正则表达式转换为 NFA。请写出详细的步骤。

1. $((\epsilon|a)^*b)^*$

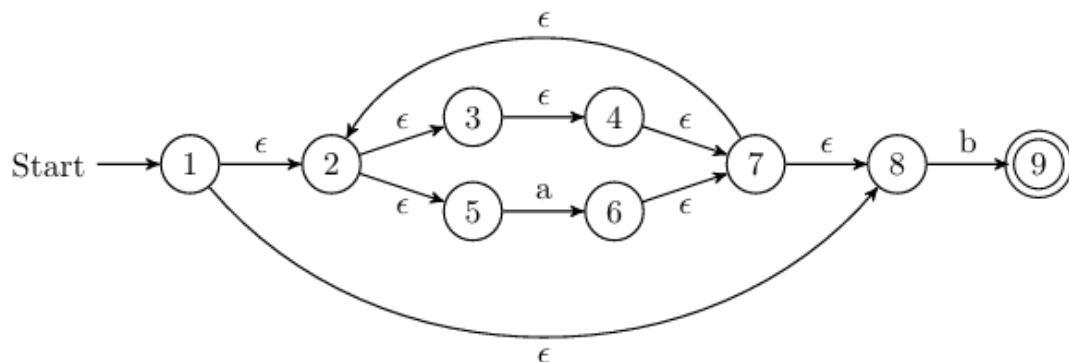
(1) Construct the NFA for $R_1 = \epsilon|a$



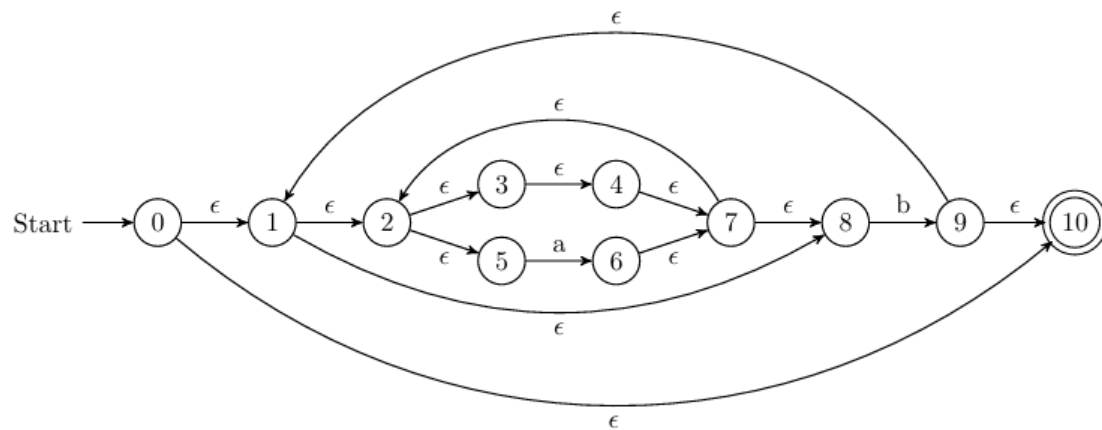
(2) Construct the NFA for $R_2 = (\epsilon|a)^* = R_1^*$



(3) Construct the NFA for $R_3 = (\epsilon|a)^*b = R_2b$

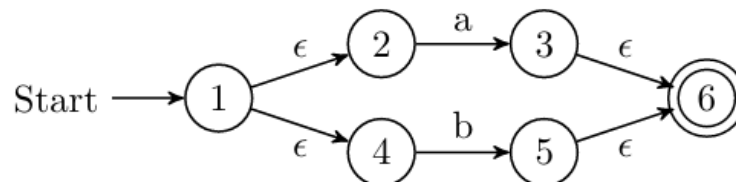


(4) Construct the NFA for $((\epsilon|a)^*b)^* = R_3^*$

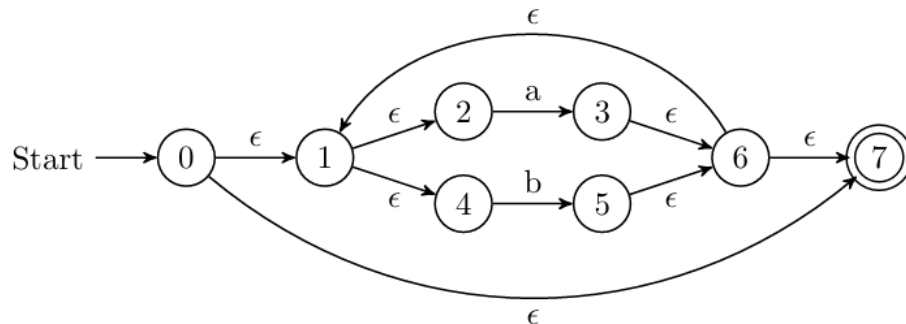


2. $(a|b)^*a(a|b)(a|b)$

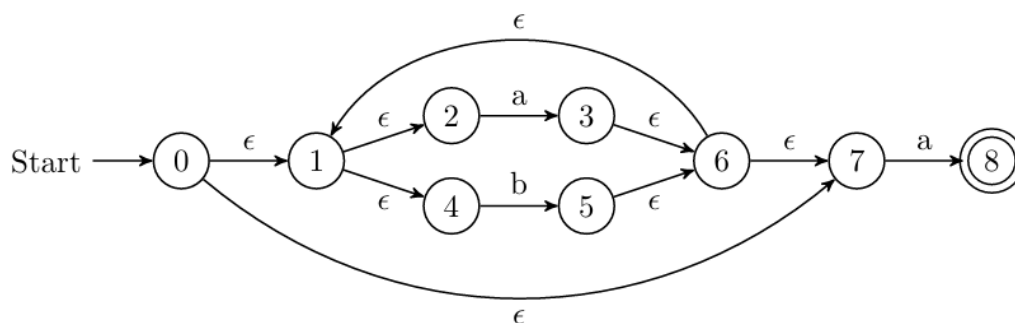
(1) Construct the NFA for $R_1 = a|b$.



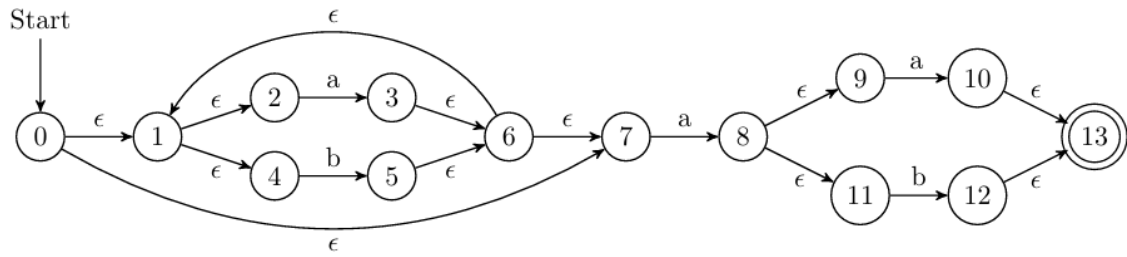
(2) Construct the NFA for $R_2 = (a|b)^* = R_1^*$.



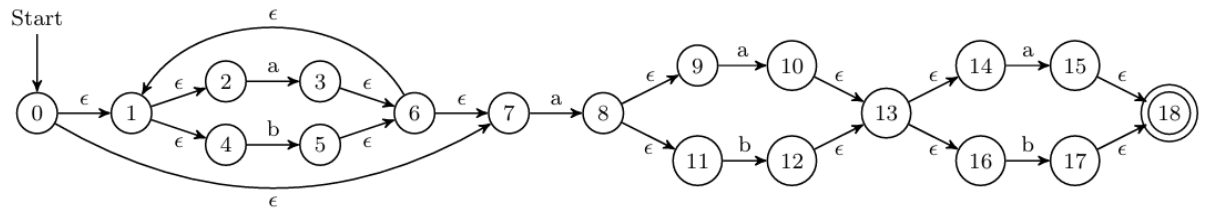
(3) Construct the NFA for $R_3 = (a|b)^*a = R_2a$.



(4) Construct the NFA for $R_4 = (a|b)^*a(a|b) = R_3R_1$.

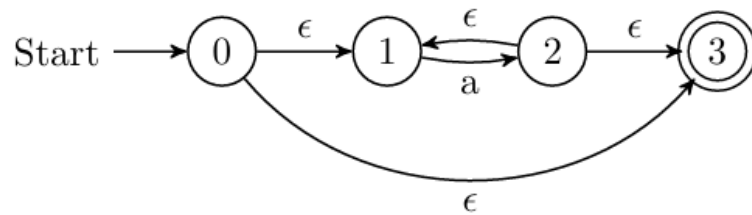


(5) Construct the NFA for $(a|b)^*a(a|b)(a|b) = R_4R_1$.

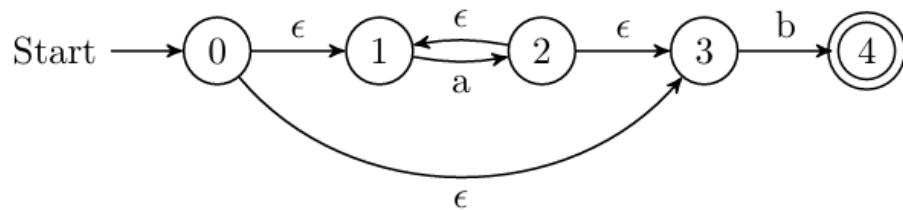


3. $a^*ba^*ba^*ba^*$

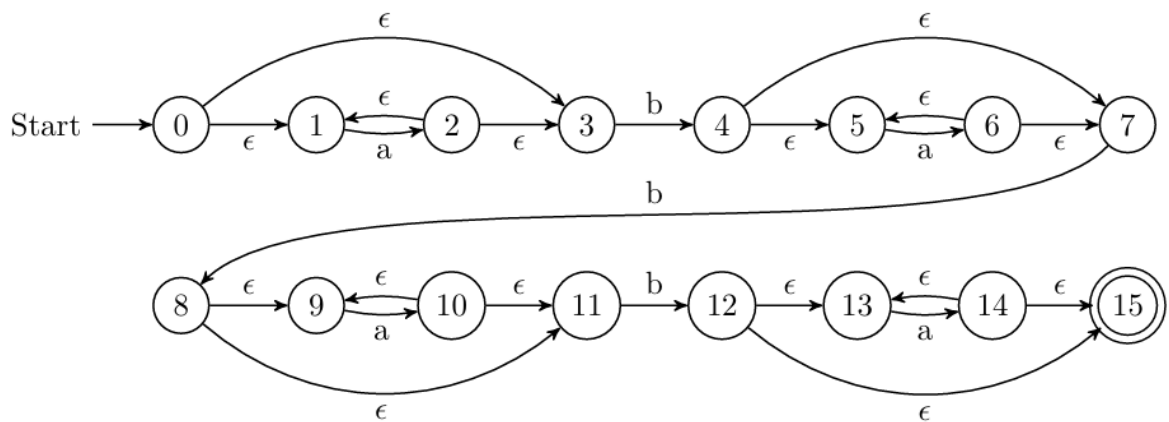
(1) Construct the NFA for $R_1 = a^*$.



(2) Construct the NFA for $R_2 = a^*b = R_1b$.



(3) Construct the NFA for $a^*ba^*ba^*ba^* = R_2R_2R_2R_1$.



Exercise 3

Convert the NFAs in Exercise2 to DFAs using the Subset Construction Algorithm (Algorithm 3.20 in the dragon book). Please put down the detailed steps. [30points in total; 10 points for each correct conversion]

请使用子集构造算法（龙书中的算法 3.20）将练习 2 中的 NFA 转换为 DFA。请写出详细步骤。[总分 30 分；每个正确转换得 10 分]

$$1. ((\epsilon|a)^*b)^*$$

$$A = \epsilon - \text{closure}(\{0\}) = \{0, 1, 2, 3, 4, 5, 7, 8, 10\}.$$

$$B = \delta_D(A, a) = \epsilon - \text{closure}(\{6\}) = \{2, 3, 4, 5, 6, 7, 8\}.$$

$$C = \delta_D(A, b) = \epsilon - \text{closure}(\{9\}) = \{1, 2, 3, 4, 5, 7, 8, 9, 10\}.$$

$$D = \delta_D(B, a) = \epsilon - \text{closure}(\{6\}) = B.$$

$$E = \delta_D(B, b) = \epsilon - \text{closure}(\{9\}) = C.$$

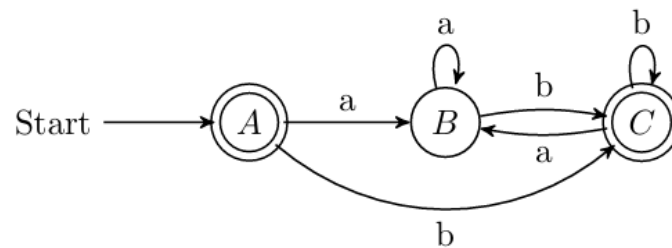
$$F = \delta_D(C, a) = \epsilon - \text{closure}(\{6\}) = B.$$

$$G = \delta_D(C, b) = \epsilon - \text{closure}(\{9\}) = C.$$

The transition table of the DFA (Starting state is A and finite states are {A, C}):

NFA STATE	DFA STATE	a	b
{0, 1, 2, 3, 4, 5, 7, 8, 10}	A	B	C
{2, 3, 4, 5, 6, 7, 8}	B	B	C
{1, 2, 3, 4, 5, 7, 8, 9, 10}	C	B	C

The transition diagram:



$$2.(a|b)^*a(a|b)(a|b)$$

$$A = \varepsilon - \text{closure}(\{0\}) = \{0, 1, 2, 4, 7\}.$$

$$B = \delta_D(A, a) = \varepsilon - \text{closure}(\{3, 8\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11\}.$$

$$C = \delta_D(A, b) = \varepsilon - \text{closure}(\{5\}) = \{1, 2, 4, 5, 6, 7\}.$$

$$D = \delta_D(B, a) = \varepsilon - \text{closure}(\{3, 8, 10\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 16\}.$$

$$E = \delta_D(B, b) = \varepsilon - \text{closure}(\{5, 12\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16\}.$$

$$F = \delta_D(C, a) = \varepsilon - \text{closure}(\{3, 8\}) = B.$$

$$G = \delta_D(C, b) = \varepsilon - \text{closure}(\{5\}) = C.$$

$$H = \delta_D(D, a) = \varepsilon - \text{closure}(\{3, 8, 10, 15\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18\}.$$

$$I = \delta_D(D, b) = \varepsilon - \text{closure}(\{5, 12, 17\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18\}.$$

$$J = \delta_D(E, a) = \varepsilon - \text{closure}(\{3, 8, 15\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 15, 18\}.$$

$$K = \delta_D(E, b) = \varepsilon - \text{closure}(\{5, 17\}) = \{1, 2, 4, 5, 6, 7, 17, 18\}.$$

$$L = \delta_D(H, a) = \varepsilon - \text{closure}(\{3, 8, 10, 15\}) = H.$$

$$M = \delta_D(H, b) = \varepsilon - \text{closure}(\{5, 12, 17\}) = I.$$

$$N = \delta_D(I, a) = \varepsilon - \text{closure}(\{3, 8, 15\}) = J.$$

$$O = \delta_D(I, b) = \varepsilon - \text{closure}(\{5, 17\}) = K.$$

$$P = \delta_D(J, a) = \varepsilon - \text{closure}(\{3, 8, 10\}) = D.$$

$$Q = \delta_D(J, b) = \varepsilon - \text{closure}(\{5, 12\}) = E.$$

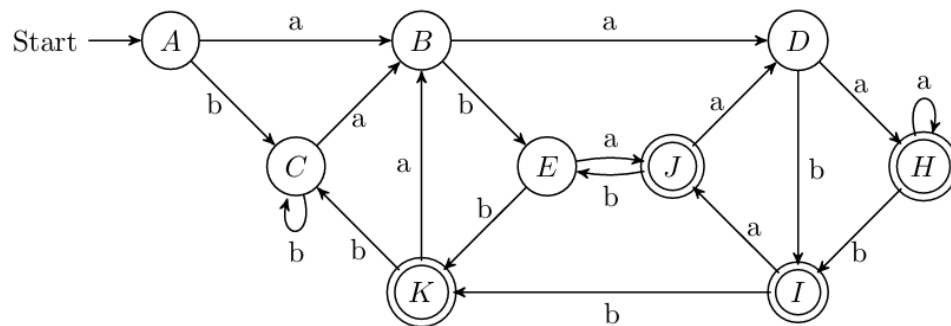
$$R = \delta_D(K, a) = \varepsilon - \text{closure}(\{3, 8\}) = B.$$

$$S = \delta_D(K, b) = \varepsilon - \text{closure}(\{5\}) = C.$$

The transition table of the DFA (Starting state is A and finite states are {H, I, J, K}):

NFA STATE	DFA STATE	a	b
$\{0, 1, 2, 4, 7\}$	A	B	C
$\{1, 2, 3, 4, 6, 7, 8, 9, 11\}$	B	D	E
$\{1, 2, 4, 5, 6, 7\}$	C	B	C
$\{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 16\}$	D	H	I
$\{1, 2, 4, 5, 6, 7, 12, 13, 14, 16\}$	E	J	K
$\{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18\}$	H	H	I
$\{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18\}$	I	J	K
$\{1, 2, 3, 4, 6, 7, 8, 9, 11, 15, 18\}$	J	D	E
$\{1, 2, 4, 5, 6, 7, 17, 18\}$	K	B	C

The transition diagram:



3. $a^*ba^*ba^*ba^*$

$A = \varepsilon - \text{closure}(\{0\}) = \{0, 1, 3\}$.

$B = \delta_D(A, a) = \varepsilon - \text{closure}(\{2\}) = \{1, 2, 3\}$.

$C = \delta_D(A, b) = \varepsilon - \text{closure}(\{4\}) = \{4, 5, 7\}$.

$D = \delta_D(B, a) = \varepsilon - \text{closure}(\{2\}) = B$.

$E = \delta_D(B, b) = \varepsilon - \text{closure}(\{4\}) = C$.

$F = \delta_D(C, a) = \varepsilon - \text{closure}(\{6\}) = \{5, 6, 7\}$.

$G = \delta_D(C, b) = \varepsilon - \text{closure}(\{8\}) = \{8, 9, 11\}$.

$H = \delta_D(F, a) = \varepsilon - \text{closure}(\{6\}) = F$.

$I = \delta_D(F, b) = \varepsilon - \text{closure}(\{8\}) = G$.

$J = \delta_D(G, a) = \varepsilon - \text{closure}(\{10\}) = \{9, 10, 11\}$.

$K = \delta_D(G, b) = \varepsilon - \text{closure}(\{12\}) = \{12, 13, 15\}$.

$L = \delta_D(J, a) = \varepsilon - \text{closure}(\{10\}) = J$.

$M = \delta_D(J, b) = \varepsilon - \text{closure}(\{12\}) = K$.

$N = \delta_D(K, a) = \varepsilon - \text{closure}(\{14\}) = \{13, 14, 15\}$.

$O = \delta_D(K, b) = \emptyset$.

$P = \delta_D(N, a) = \varepsilon - \text{closure}(\{14\}) = N$.

$Q = \delta_D(N, b) = \emptyset = O$.

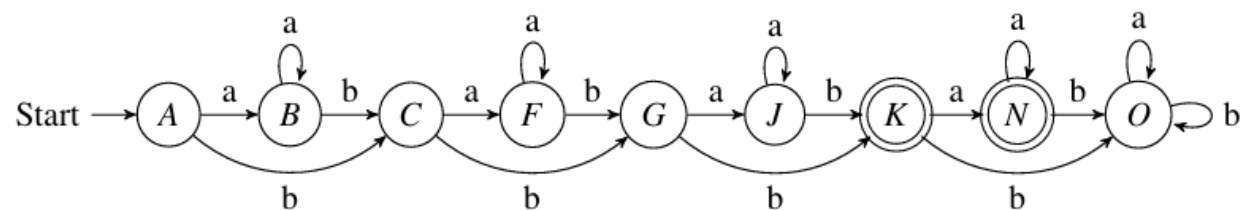
$R = \delta_D(O, a) = \emptyset = O$.

$S = \delta_D(O, b) = \emptyset = O$.

The transition table of the DFA (Strating state is A and finite states are {K, N}):

NFA STATE	DFA STATE	a	b
{0, 1, 3}	A	B	C
{1, 2, 3}	B	B	C
{4, 5, 7}	C	F	G
{5, 6, 7}	F	F	G
{8, 9, 11}	G	J	K
{9, 10, 11}	J	J	K
{12, 13, 15}	K	N	O
{13, 14, 15}	N	N	O
$\emptyset$	O	O	O

The transition diagram:



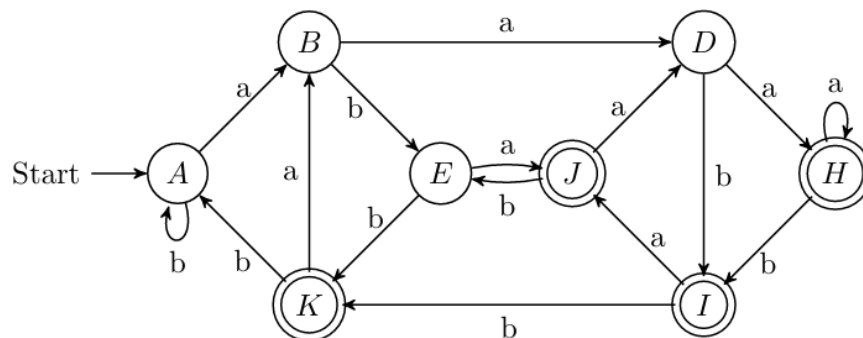
2 Optional Exercises

Exercise 1: Minimize the number of states of the DFAs you have built for regular expressions 2 and 3 in Exercise 2 using the State-Minimization Algorithm (Algorithm 3.39 in the dragon book). Please put down the detailed steps. [10 points for each correct minimization process]

使用状态最小化算法（《龙书》中算法 3.39）最小化你在练习 2 中为正则表达式 2 和 3 构建的 DFA 的状态数。请写下详细步骤。[每个正确的最小化过程 10 分]

Regular expression 2:

- (1) Initial partition: $G_1 = \{A, B, C, D, E\}$, $G_2 = \{H, I, J, K\}$.
- (2) Handling group G_1 : b splits it into two groups $G_3 = \{A, B, C\}$, $G_4 = \{D, E\}$.
- (3) Handling group G_3 : b splits it into two groups $G_5 = \{A, C\}$, $G_6 = \{B\}$.
- (4) Handling group G_4 : a splits it into two groups $G_7 = \{D\}$, $G_8 = \{E\}$.
- (5) Handling group G_2 : a splits it into two groups $G_9 = \{H, I\}$, $G_{10} = \{J, K\}$.
- (6) Handling group G_9 : b splits it into two groups $G_{11} = \{H\}$, $G_{12} = \{I\}$.
- (7) Handling group G_{10} : b splits it into two groups $G_{13} = \{J\}$, $G_{14} = \{K\}$.
- (8) Picking A, B, D, E, H, I, J, K as the representatives to construct the minimum-state DFA.



Regular expression 3:

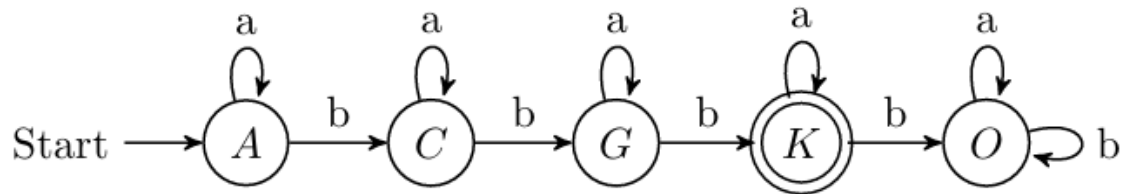
(1) Initial partition: $G_1 = \{A, B, C, F, G, J, O\}$, $G_2 = \{K, N\}$.

(2) Handling group G_1 : b splits it into two groups $G_3 = \{A, B, C, F, O\}$, $G_4 = \{G, J\}$.

(3) Handling group G_3 : b splits it into two groups $G_5 = \{A, B, O\}$, $G_6 = \{C, F\}$.

(4) Handling group G_5 : b splits it into two groups $G_7 = \{A, B\}$, $G_8 = \{O\}$.

(5) Picking A, C, G, K, O as the representatives to construct the minimum-state DFA.



Assignment 3

1 Required Exercises

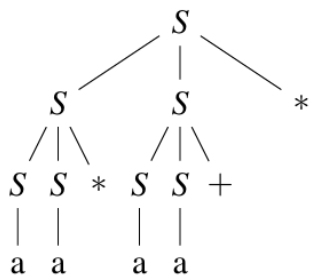
Exercise 1

Consider the following context-free grammar G :

$$S \rightarrow SS + |SS^*|a$$

1. Is the string $a + a^*a$ in $L(G)$? [10 points]
 2. Give a leftmost derivation for the string aa^*aa^+ . [10 points]
 3. Give a rightmost derivation for the string aa^*aa^+ . [10 points]
 4. Give a parse tree for the string aa^*aa^+ . [10 points]
 5. Give an equivalent grammar without immediate left recursions. [10 points]
1. 字符串 $a+a^*a$ 在 $L(G)$ 中吗?
 2. 给出该字符串的最左推导
 3. 给出该字符串的最右推导
 4. 为该字符串提供一个解析树
 5. 给出一个没有直接左递归的等价语法

1. No, it isn't. (Either a single 'a' alone, or ending with '+' or ending with '*'.)
2. $S \Rightarrow_{lm} SS \xRightarrow{*}_{lm} SS * S \xRightarrow{*}_{lm} aS * S \xRightarrow{*}_{lm} aa * S \xRightarrow{*}_{lm} aa * SS \xRightarrow{+}_{lm} aa * aS \xRightarrow{+}_{lm} aa * aa \xRightarrow{+}_{lm}$
3. $S \Rightarrow_{rm} SS \xRightarrow{*}_{rm} SSS \xRightarrow{+}_{rm} SSa \xRightarrow{+}_{rm} Saa \xRightarrow{+}_{rm} SS * aa \xRightarrow{+}_{rm} Sa * aa \xRightarrow{+}_{rm} aa * aa \xRightarrow{+}_{rm}$
- 4.



5. $S \rightarrow aS', S' \rightarrow S + S' | S * S' | \varepsilon.$

Exercise 2

Consider the following grammar G:

$$S \rightarrow aB$$

$$B \rightarrow S + B | \epsilon$$

1. Construct the predictive parsing table for G. Please put down the detailed steps, including the calculation of FIRST and FOLLOW sets. [25 points]
 2. Is the grammar LL(1)? [5 points]
 3. Can an LL(1) parser accept the input string aaaa+++? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts in the parsing table if any. [20 points]
1. 为文法 G 构造预测分析表。请写出详细步骤，包括 FIRST 和 FOLLOW 集的计算。
2. 这个语法是 LL(1) 吗？
3. LL(1) 解析器能接受输入字符串 aaaa+++ 吗？如果可以，请列出解析器的操作步骤；如果不可以，请说明原因。在解析之前，如果解析表中存在冲突，请先解决冲突。

1. (a) Calculate the FIRST and FOLLOW sets for the grammar.

$$\begin{aligned} FIRST(S) &= \{a\} & FIRST(\epsilon) &= \{\epsilon\} \\ FIRST(B) &= \{a, \epsilon\} & FOLLOW(S) &= \{+, \$\} \\ FIRST(aB) &= \{a\} & FOLLOW(B) &= \{+, \$\} \\ FIRST(S + B) &= \{a\} \end{aligned}$$

(b) Construct the predictive parsing table for the grammar.

NON-TERMINAL	INPUT SYMBOL		
	a	+	\$
S	$S \rightarrow aB$		
B	$B \rightarrow S + B$	$B \rightarrow \epsilon$	$B \rightarrow \epsilon$

2. Yes, it is.

$$\begin{aligned} FIRST(S + B) \cap FIRST(\epsilon) &= \emptyset \\ FIRST(S + B) \cap FOLLOW(B) &= \emptyset \end{aligned}$$

3. Yes, it can.

MATCHED	STACK	INPUT	ACTION
	$S\$$	$aaaa+++ \$$	
	$aB\$$	$aaaa+++ \$$	output $S \rightarrow aB$
a	$B\$$	$aaa+++ \$$	match a
a	$S + B\$$	$aaa+++ \$$	output $B \rightarrow S + B$
a	$aB + B\$$	$aaa+++ \$$	output $S \rightarrow aB$
aa	$B + B\$$	$aa+++ \$$	match a
aa	$S + B + B\$$	$aa+++ \$$	output $B \rightarrow S + B$
aa	$aB + B + B\$$	$aa+++ \$$	output $S \rightarrow aB$

<i>aaa</i>	<i>B + B + B\$</i>	<i>a + + + \$</i>	<i>match a</i>
<i>aaa</i>	<i>S + B + B + B\$</i>	<i>a + + + \$</i>	<i>output B → S + B</i>
<i>aaa</i>	<i>aB + B + B + B\$</i>	<i>a + + + \$</i>	<i>output S → aB</i>
<i>aaaa</i>	<i>B + B + B + B\$</i>	<i>+ + + \$</i>	<i>match a</i>
<i>aaaa</i>	<i>+B + B + B\$</i>	<i>+ + + \$</i>	<i>output B → ε</i>
<i>aaaa +</i>	<i>B + B + B\$</i>	<i>+ + \$</i>	<i>match +</i>
<i>aaaa +</i>	<i>+B + B\$</i>	<i>+ + \$</i>	<i>output B → ε</i>
<i>aaaa + +</i>	<i>B + B\$</i>	<i>+ \$</i>	<i>match +</i>
<i>aaaa + +</i>	<i>+B\$</i>	<i>+ \$</i>	<i>output B → ε</i>
<i>aaaa + + +</i>	<i>B\$</i>	<i>\$</i>	<i>match +</i>
<i>aaaa + + +</i>	<i>\$</i>	<i>\$</i>	<i>output B → ε</i>

Accept. The top of stack is \$.

2 Optional Exercises

Exercise 1

Consider the following context-free grammar:

短语 → 人 | 短语动词短语

动词 → 喜欢 | 讨厌

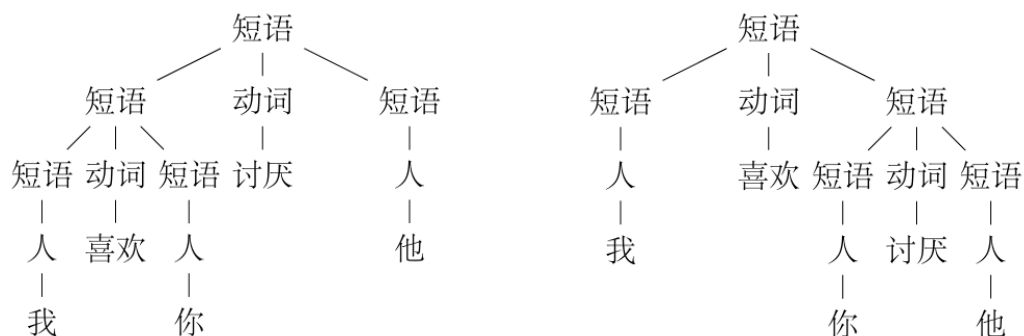
人 → 你 | 我 | 他

The grammar can produce sentences such as “我喜欢你”. Is the grammar ambiguous? If yes, please give one sentence and its multiple parse trees. If no, state the reason. [5 points for the yes/no answer and 15 points for the justification]

考虑以下无上下文文法。这个语法可以生成诸如“我喜欢你”的句子。这个语法有歧义吗？如果有，请给出一句话及其多种解析树。如果没有，请说明原因。

Yes, it is ambiguous.

e.g. 我喜欢你讨厌他



(a) Parse tree 1

(b) Parse tree 2

Figure: Two parse trees for the ambiguous sentence

Assignment 4

1 Bottom-Up Parsing Exercises

Exercise 1

(Simple LR): Consider the following grammar G:

$$S \rightarrow aB \quad B \rightarrow S+B|\epsilon$$

1. Construct the SLR(1) parsing table for G. Please put down the detailed steps, including the calculation of LR(0) item sets. [25 points]
2. Is the grammar SLR(1)? [10 points]
3. Can the SLR(1) parser accept the input string aaaa+++? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [20 points]

考虑以下文法 G:

1. 为文法 G 构造 SLR(1) 分析表。请写下详细步骤，包括 LR(0) 项目集的计算。[25 分]
2. 该文法是 SLR(1) 吗? [10 分]
3. SLR(1) 分析器能接受输入字符串 aaaa+++ 吗? 如果能，请列出分析器所做的步骤；如果不能，请说明原因。在分析之前，请解决可能存在的冲突。[20 分]

解: 1. 已知 G 为 (1) $S \rightarrow aB$
(2) $B \rightarrow SB$
(3) $B \rightarrow \epsilon$

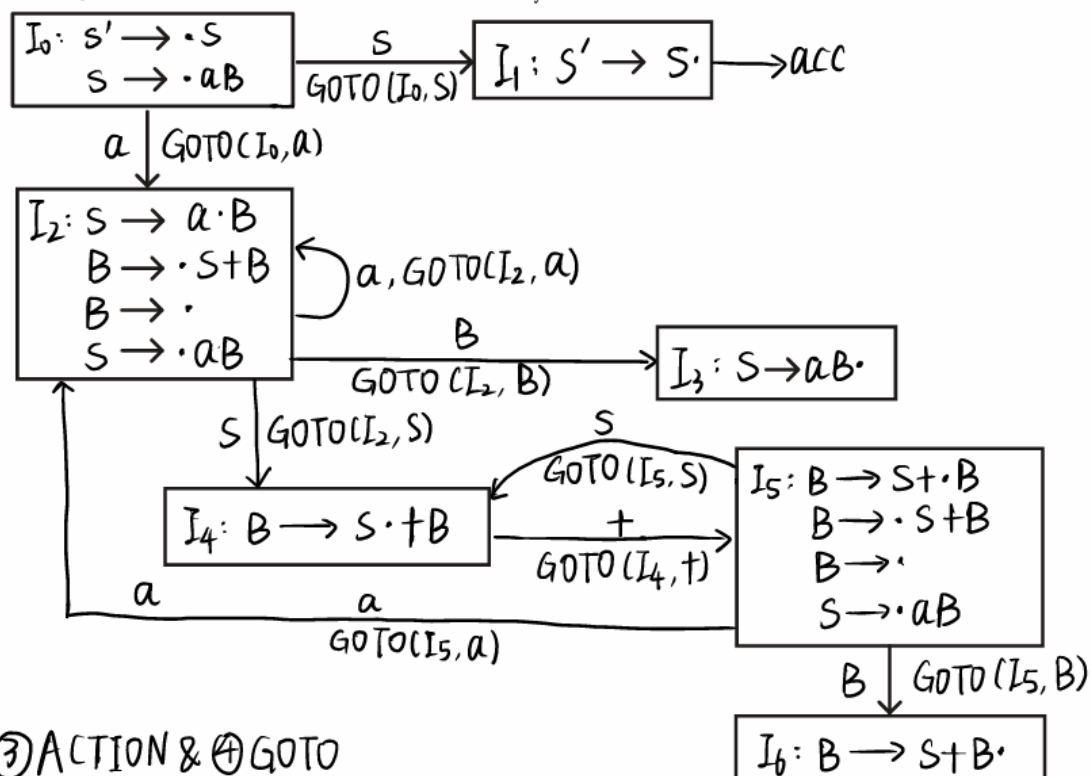
① 构造增广文法 S'

$$S' \rightarrow S$$

$$S \rightarrow aB$$

$$B \rightarrow S+B|\epsilon \quad \text{有 } \{S, a, B, +, \epsilon\}$$

② 构造 S' 的规范 LR(0) 项集族



③ ACTION & ④ GOTO

④为文法构造SLR(1)分析表

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s2	r3	r3	4	3
3		r1	r1		
4		s5			
5	s2	r3	r3	4	6
6		r2	r2		

2. Yes, it is.

3. Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	S2
(2)	0 2	a	aaa + + + \$	S2
(3)	0 2 2	aa	aa + + + \$	S2
(4)	0 2 2 2	aaa	a + + + \$	S2
(5)	0 2 2 2 2	aaaa	+ + + \$	r3 ← pop ε =0.
(6)	0 2 2 2 2 3	aaaaB	+ + + \$	r1
(7)	0 2 2 2 4	aaaS	+ + + \$	S5
(8)	0 2 2 2 4 5	aaaS +	+ + \$	r3
(9)	0 2 2 2 4 5 6	aaaS + B	+ + \$	r2
(10)	0 2 2 2 3	aaaB	+ + \$	r1
(11)	0 2 2 4	aaS	+ + \$	S5
(12)	0 2 2 4 5	aaS +	+ \$	r3
(13)	0 2 2 4 5 6	aaS + B	+ \$	r2
(14)	0 2 2 3	aaB	+ \$	r1
(15)	0 2 4	aS	+ \$	S5
(16)	0 2 4 5	aS +	\$	r3
(17)	0 2 4 5 6	aS + B	\$	r2
(18)	0 2 3	aB	\$	r1
(19)	0 1	S	\$	acc
(20)		S'		

Exercise 2

(Canonical LR): Convert Consider the grammar G in Exercise 1:

1. Construct the CLR(1) parsing table for G. Please put down the detailed steps, including the calculation of LR(1) item sets. [25 points]

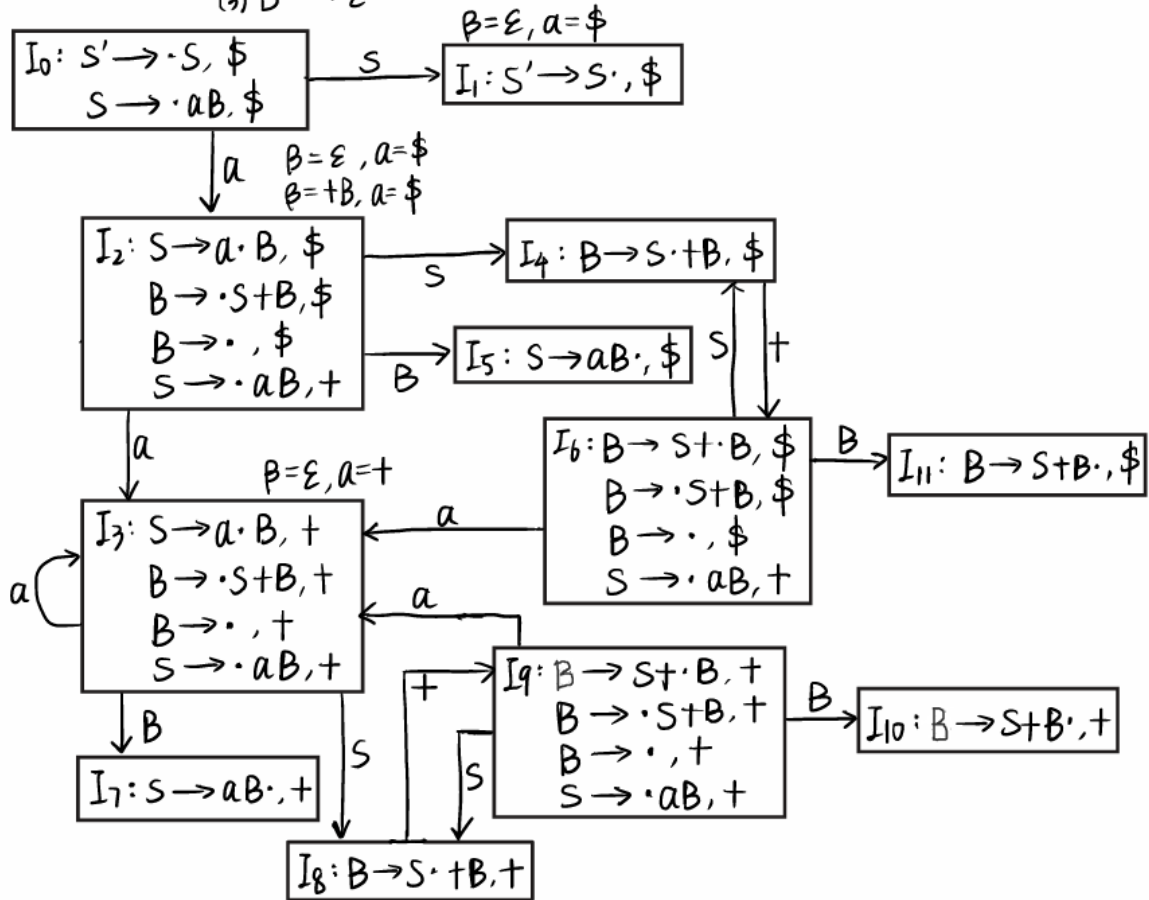
2. Can the CLR(1) parser accept the input string $aaaa++$? If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [20 points]

请考虑练习题 1 中的文法 G:

1. 构造 G 的 CLR(1) 语法分析表。请写出详细步骤，包括 LR(1) 项目集的计算。[25 分]

2. CLR(1) 分析器能否接受输入字符串 $aaaa++$? 如果可以，请列出分析器的操作步骤；如果不能，请说明原因。在解析之前，请解决可能存在的冲突。[20 分]

解: 1. 有文法为

$$\begin{aligned} (0) S' &\rightarrow S & [A \rightarrow \alpha \cdot B\beta, a] \\ (1) S &\rightarrow aB & [LR(0) \text{ item, FIRST}(B\alpha)] \\ (2) B &\rightarrow S+B & \\ (3) B &\rightarrow \varepsilon & \end{aligned}$$


STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s3	r2	r3	4	5
3	s3	r3	r2	8	7
4		s6			
5		r1	r1		
6	s3	r2	r3	4	11
7		r1	r1		
8		s9			
9	s3	r3	r2	8	10
10		r2	r2		
11		r2	r2		

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s3	r3	r3	4	5
3	s3	r3	r3	8	7
4		sb			
5		rl	rl		
6	s3	r3	r3	4	11
7		rl	rl		
8		s9			
9	s3	r3	r3	8	10
10		r2	r2		
11		r2	r2		

2. Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	S2
(2)	0 2	a	aaa + + + \$	S3
(3)	0 2 3	aa	aa + + + \$	S3
(4)	0 2 3 3	aaa	a + + + \$	S3
(5)	0 2 3 3 3	aaaa	+ + + \$	r3
(6)	0 2 3 3 3 7	aaaaB	+ + + \$	rl
(7)	0 2 3 3 8	aaaS	+ + + \$	s9
(8)	0 2 3 3 8 9	aaaS +	+ + \$	r3
(9)	0 2 3 3 8 9 10	aaaS + B	+ + \$	r2
(10)	0 2 3 3 7	aaaB	+ + \$	rl
(11)	0 2 3 8	aaS	+ + \$	s9
(12)	0 2 3 8 9	aaS +	+ \$	r3
(13)	0 2 3 8 9 10	aaS + B	+ \$	r2
(14)	0 2 3 7	aaB	+ \$	rl
(15)	0 2 4	aS	+ \$	sb
(16)	0 2 4 b	aS +	\$	r3
(17)	0 2 4 b 11	aS + B	\$	r2
(18)	0 2 5	aB	\$	rl
(19)	0 1	S	\$	acc

2 Optional Exercises

Exercise 3

(Lookahead LR): Consider the grammar G in Exercise 1:

1. Construct the LALR(1) parsing table for G. Please put down the detailed steps, including the merging of LR(1) item sets. [20 points]

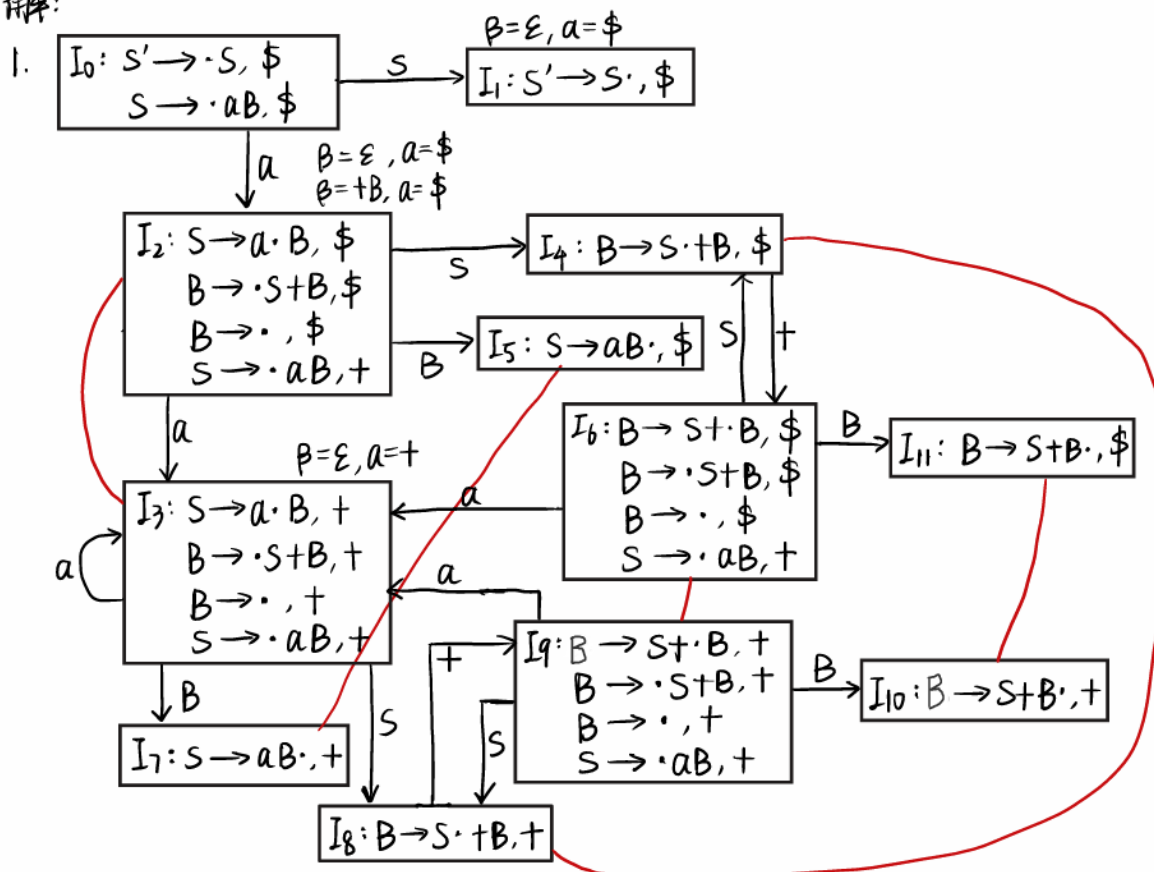
2. Can the LALR(1) parser accept the input string $aaaa++?$ If yes, please list the moves made by the parser; otherwise, state the reason. Before parsing, please resolve conflicts if any. [10 points]

(前瞻 LR): 考虑练习 1 中的文法 G:

1. 为 G 构造 LALR(1) 解析表。请写下详细步骤，包括 LR(1) 项集的合并过程。[20 分]

2. LALR(1) 解析器能否接受输入字符串 $aaaa++?$ 如果可以，请列出解析器的操作过程；如果不能，请说明原因。解析前如有冲突，请先解决。[10 分]

解:



STATE	ACTION			GOTO	
	a	+	\$	S	B
0	s2			1	
1			acc		
2	s3	r3	r3	4	5
3	s3	r3	r3	8	7
4		s6			
5		r1	r1		
6	s3	r3	r3	4	11
7		r1	r1		
8		s9			
9	s3	r3	r3	8	10
10		r2	r2		
11		r2	r2		

The *LALR(1)* parsing table for *G'*.

STATE	ACTION			GOTO	
	a	+	\$	S	B
0	S2,3			1	
1			acc		
2,3	S2,3	r3	r3	4,8	5,7
4,8		S6,9			
5,7		r1	r1		
6,9	S2,3	r3	r3	4,8	10,11
10,11		r2	r2		

2. Yes, it can.

	STACK	SYMBOLS	INPUT	ACTION
(1)	0		aaaa + + + \$	S2,3
(2)	0 2,3	a	aaa + + + \$	S2,3
(3)	0 2,3 2,3	aa	aa + + + \$	S2,3
(4)	0 2,3 2,3 2,3	aaa	a + + + \$	S2,3
(5)	0 2,3 2,3 2,3 2,3	aaaa	+ + + \$	r3
(6)	0 2,3 2,3 2,3 2,3 5,7	aaaaB	+ + + \$	r1
(7)	0 2,3 2,3 2,3 4,8	aaaS	+ + + \$	S6,9
(8)	0 2,3 2,3 2,3 4,8 6,9	aaaS +	+ + \$	r3
(9)	0 2,3 2,3 2,3 4,8 6,9 10,11	aaaS + B	+ + \$	r2
(10)	0 2,3 2,3 2,3 5,7	aaaB	+ + \$	r1
(11)	0 2,3 2,3 4,8	aaS	+ + \$	S6,9
(12)	0 2,3 2,3 4,8 6,9	aaS +	+ \$	r3
(13)	0 2,3 2,3 4,8 6,9 10,11	aaS + B	+ \$	r2
(14)	0 2,3 2,3 5,7	aaB	+ \$	r1
(15)	0 2,3 4,8	aS	+ \$	S6,9
(16)	0 2,3 4,8 6,9	aS +	\$	r3
(17)	0 2,3 4,8 6,9 10,11	aS + B	\$	r2
(18)	0 2,3 5,7	aB	\$	r1
(19)	0 1	S	\$	acc

Assignment 5

1 Required Exercises

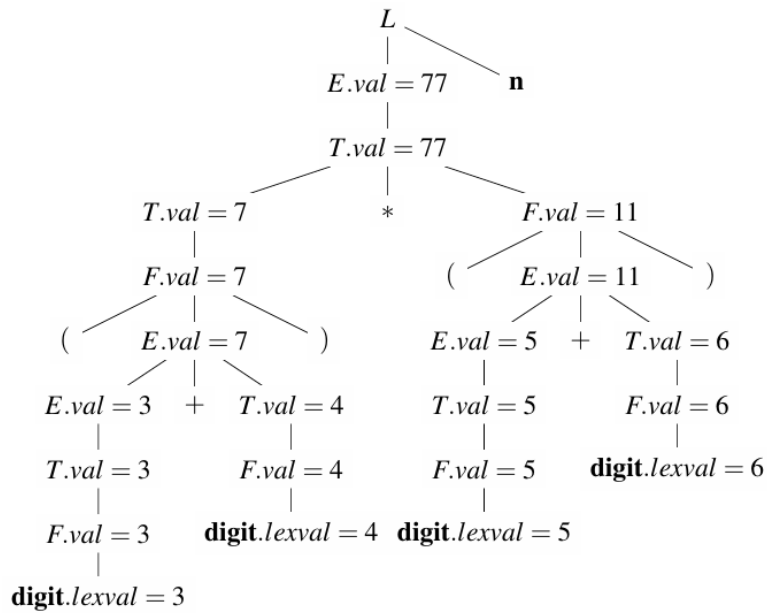
Exercise 1: For the SDD in Figure 1, give annotated parse trees for the following expressions:

1. $(3 + 4) * (5 + 6)n$ [20points]
2. $1 * 2 * 3 * (4 + 5)n$ [20 points]
3. $(9 + 8 * (7 + 6) + 5) * 4n$ [20 points]

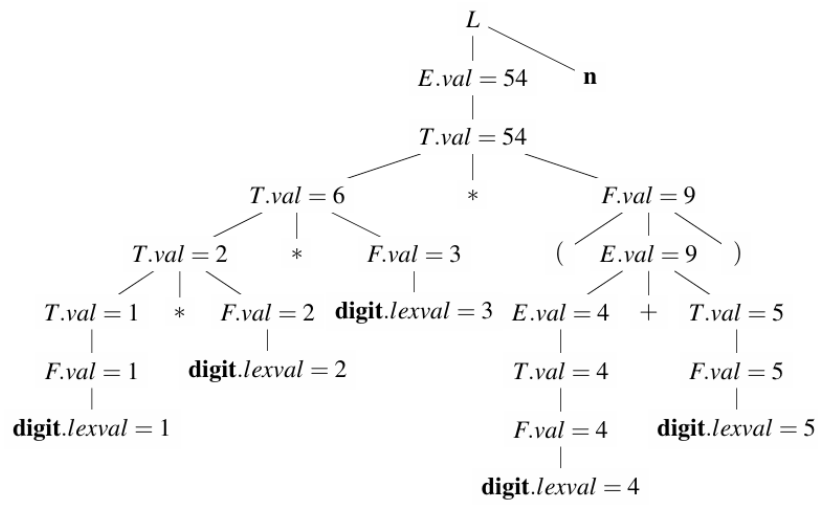
PRODUCTION	SEMANTIC RULES
1) $L \rightarrow E \mathbf{n}$	$L.val = E.val$
2) $E \rightarrow E_1 + T$	$E.val = E_1.val + T.val$
3) $E \rightarrow T$	$E.val = T.val$
4) $T \rightarrow T_1 * F$	$T.val = T_1.val \times F.val$
5) $T \rightarrow F$	$T.val = F.val$
6) $F \rightarrow (E)$	$F.val = E.val$
7) $F \rightarrow \mathbf{digit}$	$F.val = \mathbf{digit.lexval}$

Figure 1: Syntax-directed definition of a simple desk calculator

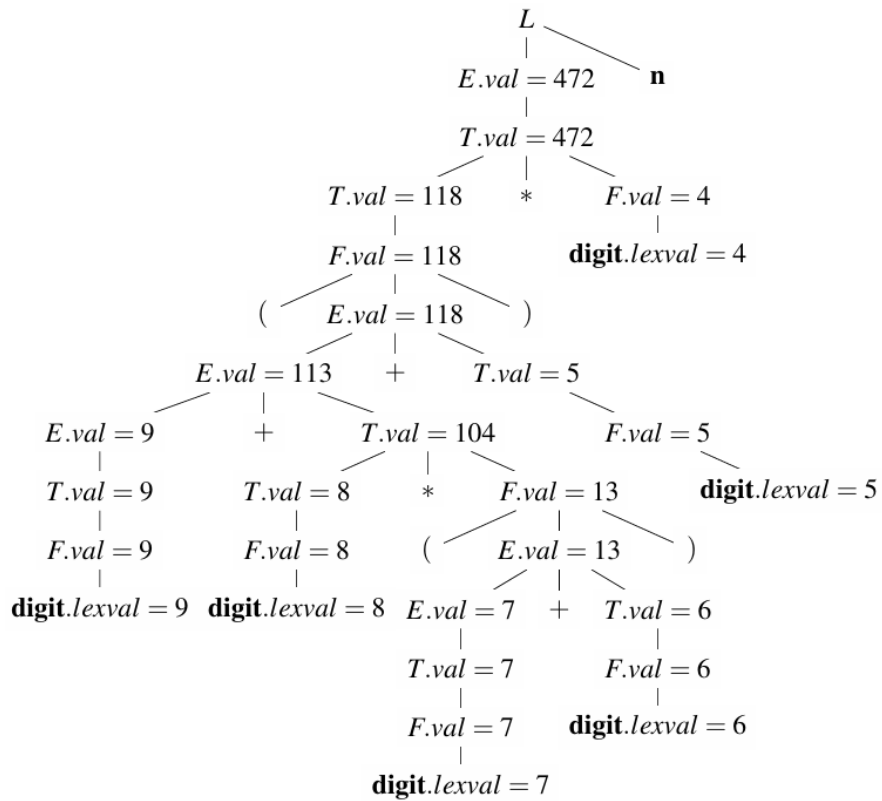
1. $(3 + 4) * (5 + 6)n$



2. $1 * 2 * 3 * (4 + 5)n$



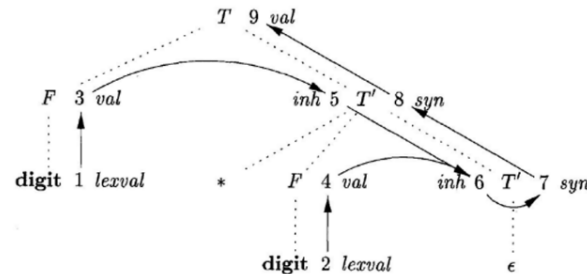
3. $(9 + 8 * (7 + 6) + 5) * 4n$



Exercise 2

What are all the topological sorts for the dependency graph of Figure 2? One sort mentioned during lecture is 1,2,3,...,9 (slide #17 of Chapter 4). [20 points]

图 2 的依赖图的所有拓扑排序有哪些？课堂上提到的一种排序是 1,2,3,...,9（第 4 章第 17 张幻灯片）。[20 分]



The productions and semantic rules for the grammar involved in figure are as follows

PRODUCTION	SEMANTIC RULES
1) $T \rightarrow F T'$	$T'.inh = F.val$ $T.val = T'.syn$
2) $T' \rightarrow * F T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3) $T' \rightarrow \epsilon$	$T'.syn = T'.inh$
4) $F \rightarrow \text{digit}$	$F.val = \text{digit}.lexval$

Based on the analysis, we can derive the sequence of attribute computation as follows, where (1, 3) indicates that step 1 must precede step 3.

(1, 3) (2, 4) (3, 5) (5, 6) (4, 6) (6, 7) (7, 8) (8, 9)

There all totally **10** different topological sorts for the dependency graph.

1	(1, 2, 3, 4, 5, 6, 7, 8, 9)
2	(1, 2, 3, 5, 4, 6, 7, 8, 9)
3	(1, 2, 4, 3, 5, 6, 7, 8, 9)
4	(1, 3, 2, 4, 5, 6, 7, 8, 9)
5	(1, 3, 2, 5, 4, 6, 7, 8, 9)
6	(1, 3, 5, 2, 4, 6, 7, 8, 9)
7	(2, 1, 3, 4, 5, 6, 7, 8, 9)
8	(2, 1, 3, 5, 4, 6, 7, 8, 9)
9	(2, 1, 4, 3, 5, 6, 7, 8, 9)
10	(2, 4, 1, 3, 5, 6, 7, 8, 9)

Exercise 3

Below is a grammar for expressions involving operator + and integer or floating-point operands. Floating-point numbers are distinguished by having a decimal point. Give an SDD to determine the type of each term T and expression E. [20 points]

$$E \rightarrow E + T \mid T$$

$$T \rightarrow num \cdot num \mid num$$

下面是一个涉及运算符+和整数或浮点操作数的表达式语法。浮点数通过有小数点来区分。请给出一个 SDD 来确定每个项 T 和表达式 E 的类型。[20 分]

PRODUCTIONS	SEMANTIC RULES
1) $E \rightarrow E1 + T$	$E.type = float \text{ if } E1.type = float \text{ or } T.type = float, \text{ else } int$
2) $E \rightarrow T$	$E.type = T.type$
3) $T \rightarrow num.num$	$T.type = float$
4) $T \rightarrow num$	$T.type = int$